

Planetary Core Energy Maintenance via Solar Wind Absorption

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PAPER_1083: Planetary Core Energy Maintenance via Solar Wind Absorption

Abstract

We derive the energy budget for planetary core maintenance through solar wind absorption. The core receives power $P_{\text{wind}} = \eta_{\text{pen}} \cdot \Phi_{\text{sw}}(d) \cdot \pi R_{\text{mag}}^2$ from wind kinetic flux, which must balance Um maintenance cost $P_{\text{Um}} = \frac{\mu_{\text{core}}^2 \omega_{\text{core}}}{\tau_{\text{Um}}}$ and radiative losses $P_{\text{rad}} = \sigma T_{\text{core}}^4 \cdot 4\pi R_{\text{core}}^2$. The net energy rate $dE/dt = P_{\text{wind}} - P_{\text{Um}} - P_{\text{rad}}$ determines whether a planet's core is in thermal equilibrium or depleting. The core SCm+UA is exclusively interactive with Ug3.

§1 Wind Power Absorbed by Core

$$P_{\text{wind}} = \eta_{\text{pen}} \cdot (1 - \eta_{\text{liquid}}) \cdot \frac{1}{2} \rho_{\text{sw}}(d) \cdot v_{\text{sw}}^3 \cdot \pi R_{\text{mag}}^2$$

§2 Um Maintenance Cost

$$P_{\text{Um}} = \frac{\mu_{\text{core}}^2 \cdot \omega_{\text{core}}}{\tau_{\text{Um}}}$$

where μ_{core} is the planetary dipole moment, ω_{core} the rotation rate, and $\tau_{\text{Um}} \sim 10^{15}$ s the magnetic diffusion timescale.

§3 Radiative Loss

$$P_{\text{rad}} = \sigma T_{\text{core}}^4 \cdot 4\pi R_{\text{core}}^2$$

§4 Core Energy Budget

$$\frac{dE_{\text{core}}}{dt} = P_{\text{wind}} - P_{\text{Um}} - P_{\text{rad}}$$

$$\tau_{\text{depletion}} = \frac{E_{\text{core}}}{|dE/dt|}$$

Equilibrium condition: $dE/dt \geq 0$ iff $P_{\text{wind}} \geq P_{\text{Um}} + P_{\text{rad}}$.

§5 Earth Example

Parameter	Value	Source
d	1.0 AU	Orbital distance
R_{mag}	6.4×10^7 m	Earth magnetosphere

R_{core} | $3.486 \times 10^6 \text{ m}$ | Inner+outer core |
 T_{core} | 5700 K | Core temperature |
 μ_{core} | $8 \times 10^{22} \text{ A} \cdot \text{m}^2$ | Geomagnetic dipole |
 P_{wind} | $\sim 10^{10} \text{ W}$ | Wind power to core |
 P_{U_m} | $\sim 10^{15} \text{ W}$ | Magnetic maintenance |
 P_{rad} | $\sim 10^{19} \text{ W}$ | Radiative loss |
 dE/dt | < 0 | Net depletion |

§6 Ug3 Exclusivity

Core SCm+UA interacts exclusively with Ug3 (magnetic string rotation), not with Ug1 (dipole) or Ug2 (charge-reactivity). This means:

- Core energy maintenance depends on Ug3 string coupling distance
- Beyond the frost line, Ug3 coupling weakens and wind power dominates (PAPER_1079)
- Core dynamo longevity correlates with Ug3 string tension retention

§7 Planetary Comparison

Planet	P_{wind}	P_{U_m}	P_{rad}	Regime
Earth	$\sim 10^{10} \text{ W}$	$\sim 10^{15} \text{ W}$	$\sim 10^{19} \text{ W}$	Depleting (slowly)
Jupiter	$\sim 10^{14} \text{ W}$	$\sim 10^{18} \text{ W}$	$\sim 10^{21} \text{ W}$	Depleting
Neptune	$\sim 10^{11} \text{ W}$	$\sim 10^{13} \text{ W}$	$\sim 10^{17} \text{ W}$	Depleting
Mars	$\sim 10^6 \text{ W}$	0 W	$\sim 10^{16} \text{ W}$	Dead dynamo

References

- PAPER_1078: Solar Wind Flux Partitioning Budget
- PAPER_1079: Frozen Planet Solar Wind Power
- PAPER_413: Ug3CCWCWDifferentialRotationSCmPlanetaryCoreCalculator
- Star-Magic.txt: §5 Core Energy Maintenance

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic